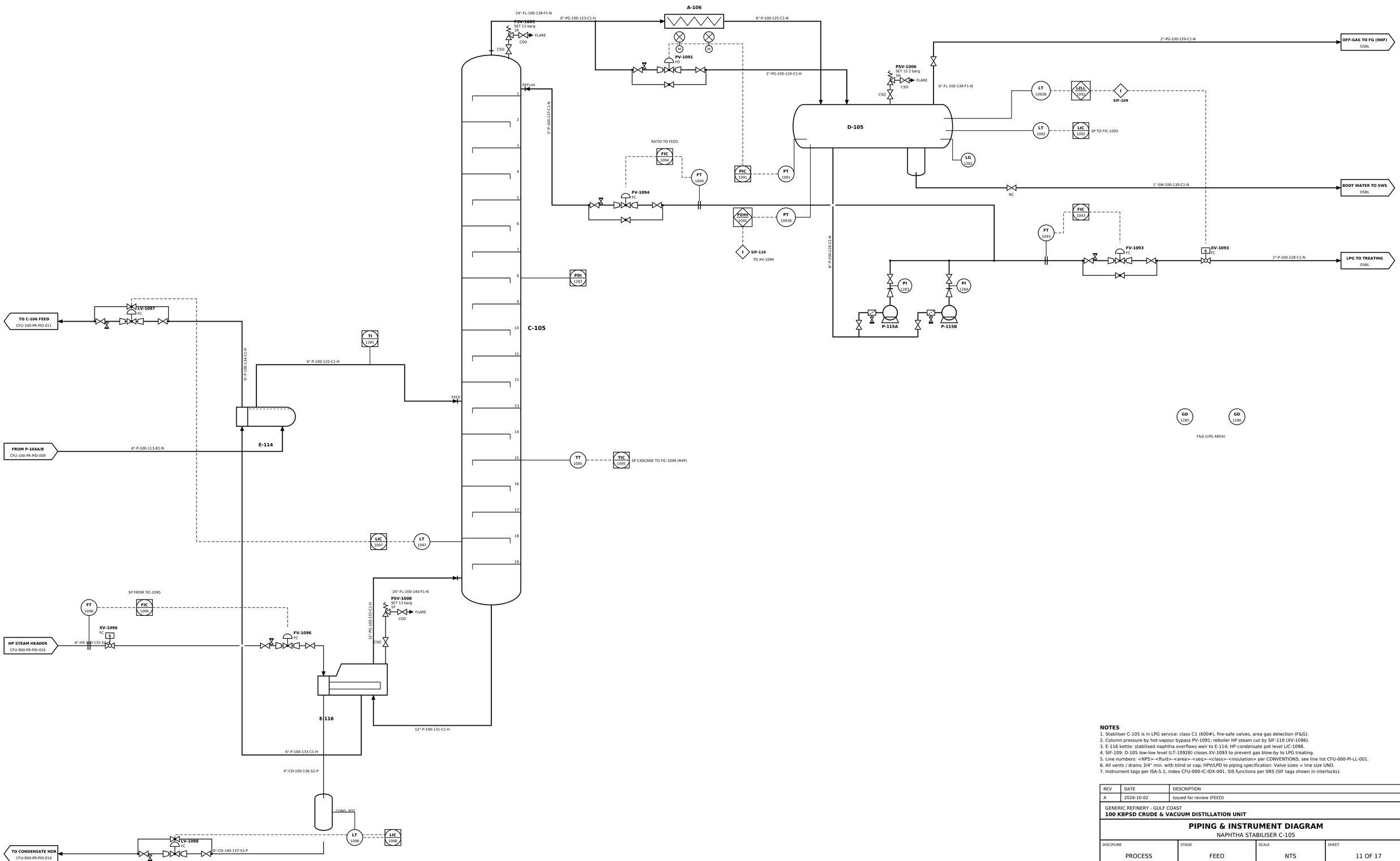
C-105	E-114	E-116	A-106	D-105	P-115A/B
Naphtha stabiliser (debutaniser) ID 2.1 m x 18.0 m T/T DESIGN 13.0 barg / 235 C; CA 3 mm 19 valve trays, feed tray 13 MAT'L CS (HIC-resistant plate, NACE MR0103)	Stabiliser feed / bottoms 86 m2; 1 shell(s) DESIGN 20 / 20 barg / 235 C; CA 3 mm DUTY 3.38 MW SHELL: Unstab. naphtha / TUBE: Stabiliser bottoms MAT'L CS	Stabiliser reboiler (HP steam) 242 m2; 1 shell(s) DESIGN 15 / 45 barg / 430 C; CA 3 mm DUTY 10.83 MW SHELL: Naphtha / TUBE: HP steam MAT'L CS	Stabiliser overhead condenser 214 m2 bare; 1 bay(s) 6 m x 12 m, 2 fans DESIGN 16.7 barg / 110 C DUTY 2.14 MW; 2 x 11 kW fans MAT'L CS tubes, Al fins (A-101: CS + inlet ferrules, Ti optional)	Stabiliser reflux drum ID 1.7 m x 5.5 m T/T; boot ID 0.6 m x 1.5 m DESIGN 15.2 barg / 95 C; CA 3 mm MAT'L CS (HIC-resistant)	Stabiliser reflux / LPG RATED 39 m3/h @ 154 m, dP 8 bar MOTOR 18.5 kW; API 682 Plan 11/52 DES.: T 75 C; MAT'L S-6 (CS/12Cr)



NOTES

1. Stabiliser C-105 is in LPG service: class C1 (600#), fire-safe valves, area gas detection (F&G).
2. Column pressure by hot-vapour bypass PV-1091; reboiler HP steam cut by SIF-110 (XV-1096).
3. E-116 kettle: stabilised naphtha overflows weir to E-114; HP condensate pot level LIC-1098.
4. SIF-109: D-105 low-low level (LT-1092B) closes XV-1093 to prevent gas blow-by to LPG treating.
5. Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
6. All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
7. Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - GULF COAST

100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

PIPING & INSTRUMENT DIAGRAM

NAPHTHA STABILISER C-105

DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	11 OF 17

DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02

DRAWING NUMBER **CFU-100-PR-PID-010** REV **A**